Foundations

- A permutation of size n is an ordering of $1, 2, \dots n$ Ex. $\{ 123, 321, \dots \}$

  Lemma: there are $n!$ permutations of length n

  Proof: n ways for 1st choice ...

- A binary string of length n is an n-tuple made of 0s and 1s

  Ex. All binary strings of $n=2$ are $00, 01, 10, 11$

  Lemma: there are 2^n binary strings

  Proof: every pos has 2 choices, n choices $\to 2^n$

Bijections

Idea, use ideas we already know to define new ideas

Let A, B be 2 sets.

Then $f: A \mapsto B$ is

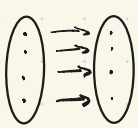
injective if $a \neq a' \Rightarrow f(a) \neq f(a')$ $\forall a, a' \in A$ or $f(a) = f(a') \Rightarrow a = a'$

surjective if $b \in B$, $\exists\ a \in A$ s.t. $f(a) = b$

bijective both injective and surjective

Key: If A and B are finite and in bijection, $|A| = |B|$

ie



injective surjective

$f: A \to B$ is a bijection iff $\exists$ a $g: B \to A$ s.t. $f(g(b)) = b$ $\forall b \in B$ & $g(f(a)) = a$ $\forall a \in A$

"well-defined" inverse

Proving whether f is a bijection we can 1. show injective + surjective

2. find inverse.

Subsets

A subset of $[n] = \{1, \dots, n\}$ is a collection of some elements in $[n]$, possibly empty

The subset of $[0] = \emptyset$ or β (empty set)

Lemma: $n \in \mathbb{N}$ $\exists$ 2^n subsets of $[n]$

Proof: every element is either picked or not $\to 2^n$

Proof (Bijection): Fix n

let A = subsets of $[n]$

let B = binary strings of length n

let $f: A \to B$ be defined by

$f(s) = \sigma_1 \sigma_2 \cdots \sigma_n$, where $\sigma_i \begin{cases} 0 & \text{if } i \notin S \\ 1 & \text{if } i \in S \end{cases}$

we need to show

① f is well-defined (need to show $a \in A \to f(a) = b$, $b \in B$)

② f is injective

③ f is surjective

1. f is always length n, and is always either 0 or 1

2. If $f(S) = f(T)$, then $i \in S \iff i \in T \Rightarrow S = T$

3. If $\sigma_1 \sigma_2 \cdots \sigma_n \in B$ let $S = \{i : \sigma_i = 1\}$

   Then $S \in A$ and $f(S) = \sigma_1 \cdots \sigma_n$ by definition

   $\therefore |A| = |B| = 2^n$

Lemma: There are $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ subsets of $[n]$ with k elements ($n \in \mathbb{N}$, $0 \le k \le n$)

Proof: There are $n \times (n-1) \times (n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}$ ways to pick a k-length tuple using distinct elements of $[n]$

A subset of size k is a tuple of length k where the order does not matter. Since there are $k!$ ways to order the tuple of length k, there are

$\frac{1}{k!} \cdot \frac{n!}{(n-k)!} = \binom{n}{k}$ subsets of $[n]$ of length k.

Multiset

- A multiset of size n with t types is a set with n elements in $\{1, \dots t\}$ that can repeat
- A sequence $(m_1, \dots, m_t) \in \mathbb{N}^t$ with $m_1 + \dots + m_t = n$

(The two can be related by letting m_i in B be the number of elements of type i.)

Theorem: If there are $n, t \in \mathbb{N}$ and $t \geq 1$ then there are

$$\binom{n+t-1}{t-1}$$

multisets of size n, t types.

Idea for proof: $m_1 + m_2 + m_3 = 5$ $\quad \bullet \bullet \bullet \bullet \bullet$ } 5 items

$1 + 2 + 2 = 5 \rightarrow \quad \bullet \mid \bullet \bullet \mid \bullet \bullet$

we can move lines around.

In general, we have n balls, $t-1$ lines to divide.
We pick the $t-1$ spots for lines, in the $n+t-1$ positions.

$$\binom{n+t-1}{t-1}$$

Combinatorial Proofs

Bijections can give combinatorial proofs of identities (LHS = RHS)

Steps:

① Find sets $|A|$ and $|B|$ s.t. $LHS = |A|$, $RHS = |B|$

② Prove A and B are in bijective

③ Conclude $LHS = |A| = |B| = RHS$

Special Case: count 1 set in 2 ways

Ex) Give a combinatorial Proof that $\binom{n}{k} = \binom{n}{n-k}$ $n, k \in \mathbb{N}$, $0 \le k \le n$

Proof:

let $A = \{$ subsets of $[n]$ of size $k\}$

$B = \{$ subsets of $[n]$ of size $n-k\}$

by binomial coeff, $|A| = \binom{n}{k}$, $|B| = \binom{n}{n-k}$

Define $f: A \to B$ by $f(S) = [n] \setminus S$ (the full set, $[n]$ subtracted by S)

① If $|S| = k$ then $|f(S)| = |[n] \setminus S| = n-k$

Hence, it is well-defined.

② $f(f(S)) = [n] \setminus ([n] \setminus S) = S$

So f is its own inverse

Thus f is a bijection and $\binom{n}{k} = |A| = |B| = \binom{n}{n-k}$

Ex) Prove $2^n = \sum_{k=0}^{n} \binom{n}{k}$

can be subsets — inductive of subsets

can be binary strings

Proof: let $A = \{\text{subsets of } [n]\}$, so $|A| = 2^n$

we can write A as a disjoint union of: {Disjoint, want unions + 0 not double count

$A = A_0 \cup A_1 \cup A_2 \dots \cup A_n$, where $A_k = \{\text{subsets of } [n] \text{ with size } k\}$

As it is disjoint,

$$2^n = |A| = \left| \bigcup_{k=0}^{n} A_k \right|$$

$$= \sum_{k=0}^{n} |A_k|$$

$$= \sum_{k=0}^{n} \binom{n}{k}$$

Common Objects in Proofs:

- subsets
- Permutations
- (binary) strings
- multisets
- lattice Path

A lattice path consists of a sequence of steps that start and end at specific points in $\mathbb{Z}^1$

Ex (Lattice paths): Find the number of lattice paths in $\mathbb{Z}^2$ that start at $(0,0)$ and end at $(a,b) \in \mathbb{N}^2$ using the steps $N = (0,1)$ $E = (1,0)$

Sol:

A walk ending at (a,b) must have a E steps, and b N steps $= a+b$ steps

Thus, to pick a E-steps, the remaining b steps are set uniquely, hence there are $\binom{a+b}{a}$ paths from $(0,0) \to (a,b)$

General framework for Enumeration

A combinatorial class is a set A together with a size or weight function $w: A \to \mathbb{N}$ such that there is a finite set of objects of any size.

$$A_n = w^{-1}(n) = \{a \in A : w(a) = n\}$$

is finite $\forall\, n \in \mathbb{N}$. Typically, $w(\alpha) = |\alpha|$

} does there need to be every $|\alpha| \in \mathbb{N}$
No, just finite for a weight.

Ex) Binary strings

$B = \{\varepsilon, 0, 1, 00, 01, 10, 11, \dots\}$

↙ sets of objects B with size functions (w, or $|B|$)

- If $|s|$ = length(s) then $(B, |\cdot|_B)$ is a combinatorial class
- If $|s|$ = # of 0s in s then is not a combinatorial class.

$|B_0| = \infty$ (consider $B = \{\varepsilon, 1, 11, \dots\}$)

↙ we consider sets for weights because we may have previously considered bijections $1 \to n$, but this is only.

A Bijection between classes $(A, |\cdot|_A)$ and $(B, |\cdot|_B)$ is a bijection $f: A \to B$ that respects size

$$|a|_A = |f(a)|_B \quad \forall\, a \in A.$$

The counting sequence of a class $(A, |\cdot|)$ is defined by

$a_n = |A_n|$ = # of objects of size n

ie, B = binary strings then $b_n = 2^n$

Generating Function (generating series)

The generating function of a sequence (a_n) is the series $\sum_{n\geq 0} a_n z^n = a_0 + a_1 z + a_2 z^2 \dots$

The GF of a class $(A, |\cdot|)$ is the GF of its counting sequence

Ex. The GF for the class $\mathcal{B}$ of binary strings is

$$B(z) = \sum_{n\geq 0} b_n z^n = \sum_{n\geq 0} 2^n z^n$$

This is known as a formal series

Ex $\sum_{n\geq 0} 2^n z^n$ converges $|z| < \frac{1}{2}$

$\sum_{n\geq 0} n!\, z^n$ converges only $z=0$

Formal series:

we define addition and multiplication

- $\sum_{n\geq 0} a_n x^n + \sum_{n\geq 0} b_n x^n = \sum_{n\geq 0} (a_n + b_n) x^n$
- $\left(\sum_{n\geq 0} a_n x^n\right)\left(\sum_{n\geq 0} b_n x^n\right) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n$

we can show the usual properties still hold formally

we define $\mathbb{Q}[[x]]$ the ring of formal power series (infinite series polynomials w/ rational coeff)

$\mathbb{Q}[x]$ → polynomials

Ex). Consider $F(x) = 1-x$. Does there exist some $G(x) = \sum_{n\geq 0} g_n x^n$ s.t. $F(x)\cdot G(x) = 1$

Let $G(x) = \sum_{n\geq 0} x^n$, then $F(x)\cdot G(x) = (1-x)\left(\sum_{n\geq 0} x^n\right)$

$$= \sum_{n\geq 0} x^n - x\sum_{n\geq 0} x^n$$

$$= 1 + \sum_{n\geq 1} x^n - \sum_{n\geq 1} x^n$$

$$= 1$$

So $\frac{1}{F(x)} = G(x) = \frac{1}{1-x}$ as formal series

Ex) $F(x) = x+x^2$

$$\frac{1}{F(x)} = \frac{1}{x+x^2} = \frac{1}{x}\cdot\frac{1}{1+x}$$

$$= \frac{1}{x}\cdot\sum_{n\geq 0}(-1)^n x^n$$

$G(x)F(x)$ is not possible as constant term $= \frac{1}{x}$

Lemma: $\frac{1}{F(x)}$ exists in $\mathbb{Q}[[x]] \iff f_0 \neq 0$

Lemma: $F(G(x))$ exists $\iff$ either

1) F is polynomial
2) $g_0 = 0$

Consider $F(G(x)) = \sum f_n G(x)^n$ ← if not constant coeff not working

Ex) Find the GF for m_n = # of ways to make change for n cents using $1^{¢}$, $5^{¢}$, $10^{¢}$, $25^{¢}$ coins.

Let $M = \{(a,b,c,d) \in \mathbb{N}^4\}$ and define $|(a,b,c,d)| = a + 5b + 10c + 25d$

tells you value. (counting sequence counts objects)

Then the Generating Function of M is

$$m(x) = \sum_{\sigma \in M} x^{|\sigma|}$$

$$= \sum_{(a,b,c,d) \in M} x^{a+5b+10c+25d}$$

more formally, we have $F(x) = \frac{1}{1-x}$, $G(x) = x^5$

as $G(x)$ is non-constant $F(G(x))$ is ok. by

$$= \left(\sum_{a \geq 0} x^a\right)\left(\sum_{b \geq 0} x^{5b}\right)\left(\sum_{c \geq 0} x^{10c}\right)\left(\sum_{d \geq 0} x^{25d}\right)$$

$$= \left(\frac{1}{1-x}\right)\left(\frac{1}{1-x^5}\right)\left(\frac{1}{1-x^{10}}\right)\left(\frac{1}{1-x^{25}}\right)$$

Computing:

$$m(x) = 1 + x + x^2 + \cdots + 242\,x^{100} + \cdots$$

242 ways to make change for \$1

asymptotically, $m_n \sim n^3/7500$

Ex) the GF of number of ways to roll 2 standard dice is

$$D(x) = \left(x + x^2 + x^3 + x^4 + x^5 + x^6\right)^2$$

$$= \left(x + 2x^2 + 2x^3 + x^4\right)\left(x + x^3 + x^4 + x^5 + x^6 + x^8\right)$$

sum of coeffs $= 6$

If you take 2 dice with faces

$(1, 2, 2, 3, 3, 4)$

$(1, 3, 4, 5, 6, 8)$

the probability of rolling any number is the same

Recall:

- GFs $\quad C(x) = \sum_{n \geq 0} c_n x^n = \sum_{a \in \ell} x^{|a|}, \quad \ell = \{\dots\}$

- $\frac{1}{F(x)}$ exists as formal series $\iff F(0) \neq 0$

- $G\left(F(x)\right)$ exists $\iff G$ polynomial or $F(0) = 0$

Binomial Theorem

If $n \in \mathbb{N}$, then $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$

Negative Binomial Theorem

If t is a positive integer, then

$$\frac{1}{(1-x)^t} = \sum_{n \geq 0} \binom{n+t-1}{t-1} x^n$$

proof:

The RHS is the GF for the number of multisets with t types

Thus, $\sum_{n \geq 0} \binom{n+t-1}{t-1} x^n = \sum_{(m_1, \ldots, m_t) \in \mathbb{N}^t} x^{m_1 + \cdots + m_t} = \left(\sum_{m_1 \geq 0} x^{m_1}\right) \cdots \left(\sum_{m_t \geq 0} x^{m_t}\right)$

GF def'n, $\sum_{n \geq 0} c_n x^n = \frac{1}{(1-x)^t}$

Generalized Binomial Theorem

Let $\binom{\alpha}{k} = \frac{\alpha!}{(\alpha-k)!\,k!} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}$, for any $k \in \mathbb{N}$ and $\alpha \in \mathbb{C}$

For any $\alpha \in \mathbb{C}$, $(1+x)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} x^n$

Note, from the negative Binomial Theorem, $\binom{n+t-1}{t-1} = (-1)^n \binom{-t}{n}$

we write $[x^n]\left(\sum_{k\geq 0} c_k x^k\right) = c_n$

ie, $[x^0]F(x) = f_0 = F(0)$

Rules for coefficient extraction

- $[x^n]\,\lambda A(x) = \lambda [x^n] A(x)$
- $[x^n](A(x)+B(x)) = [x^n]A(x) + [x^n]B(x)$
- $[x^n]F(ax) = a^n [x^n] F(x)$ $\quad \begin{cases} F(x) = \sum f_n x^n \\ F(ax) = \sum f_n a^n x^n \end{cases}$
- $[x^n]\,x^k F(x) = [x^{n-k}]\,F(x)$, where $n-k = 0$ if $k > n$

Ex) Find $[x^n]\dfrac{1-\sqrt{1-4x}}{2x}$ (catalan series, is in bijection w/ a lot of objects)

$$[x^n]\frac{1-\sqrt{1-4x}}{2x} = \frac{1}{2}[x^{n+1}]\left(1-\sqrt{1-4x}\right)$$

$$= -\frac{1}{2}[x^{n+1}]\sqrt{1-4x}$$

$$= -\frac{1}{2}(-4)^{n+1}[x^{n+1}](1+x)^{1/2}$$

$$= -\frac{1}{2}(-4)^{n+1}\binom{1/2}{n+1}$$

$$= -\frac{1}{2}(-4)^{n+1}\frac{(1/2)(1/2-1)\cdots(1/2-n)}{(n+1)!}$$

$$= 2^n(-2)^n\frac{(1/2-1)(1/2-2)\cdots(1/2-n)}{(n+1)!}$$

$$= 2^n\frac{(1)(3)(5)\cdots(2n-1)}{(n+1)!}\cdot\frac{n!}{n!}$$

(all products; $n!\,2^n$ gives all n even products)

$$= \frac{(2n)!}{n!\,n!}\cdot\frac{1}{(n+1)}$$

$$= \binom{2n}{n}\cdot\frac{1}{(n+1)}$$

Combinatorial classes

Idea: Build complicated classes out of simple ones, and relate to GFs

Two Base Cases:

① Atomic class

$\mathcal{Z} = \{\bullet\}$ single object of size 1

The GF of this class is Z

② Neutral class $\mathcal{E} = \{\varepsilon\}$ = one object of size 0

The GF of this class is 1

Constructions

① Sum

Let $(A, |\cdot|_A)$ and $(B, |\cdot|_B)$ be 2 combinatorial classes

The sum $C = A+B$ is the class whose objects are the disjoint union of the objects in A and B

Ex) Let $\mathcal{Z} = \{\bullet\}$ be the atomic class. Consider $\mathcal{Z}+\mathcal{Z} = \{\bullet, \bullet\}$ be the class of 2 distinct atoms of size 1.

The size function is $|\sigma|_C = \begin{cases} |\sigma|_A : \sigma \in A \\ |\sigma|_B : \sigma \in B \end{cases} \quad \forall\ \sigma \in C$

Lemma: If $C = A+B$ then $C(z) = A(z) + B(z)$

Proof: An object of size n in C is an object of size n in A or B since we take the disjoint union, thus counting sequences satisfy

$$c_n = a_n + b_n$$

Thus,

$$\begin{aligned} C(z) &= \sum_{n \geq 0} c_n z^n \\ &= \sum_{n \geq 0} (a_n + b_n) z^n \\ &= \sum_{n \geq 0} a_n z^n + \sum_{n \geq 0} b_n z^n \\ &= A(z) + B(z) \end{aligned}$$

② Product

The Product $C = A \times B$ is the class whose objects are all pairs

$$\{(\alpha, \beta) : \alpha \in A, \beta \in B\}$$

The size is defined by

$$|(\alpha, \beta)|_C = |\alpha|_A + |\beta|_B$$

Lemma:

If $C = A \times B$ then $C(z) = A(z) B(z)$

$$\begin{aligned} C(z) &= \sum_{(\alpha,\beta) \in C} z^{|(\alpha,\beta)|} \\ &= \sum_{(\alpha,\beta) \in C} z^{|\alpha|_A + |\beta|_B} \\ &= \left(\sum_{\alpha \in A} z^{|\alpha|_A} \right) \left(\sum_{\beta \in B} z^{|\beta|_B} \right) = A(z) B(z) \end{aligned}$$

③ Powers

For any positive integer k,

$$A^k = \underbrace{A \times A \times \dots A}_{k \text{ times}} = \{(\alpha_1, \dots, \alpha_k) : \alpha \in A\}$$

with GF $A(z)^k$

④ Sequence

Let A be a combinatorial class with no objects of size 0

The sequence class of A is

$$\begin{aligned} SEQ(A) &= \mathcal{E} + A + A^2 + \dots \\ &= \bigcup_{k \geq 0} \{(\alpha_1, \dots, \alpha_k) : \alpha \in A\} \end{aligned}$$

Note: If A has something of size 0, then $SEQ(A)$ would have an infinite set of objects of size 0.

Lemma: If $C = SEQ(A)$ then $C(z) = 1 + A(z) + A(z)^2 + \dots$

$$= \frac{1}{1 - A(z)}$$

Ex) Let B = class of binary strings

$$B = SEQ(\mathcal{Z}_0 + \mathcal{Z}_1) \quad \text{atoms of sets } \mathcal{Z} = \{0\} \text{ and } \mathcal{Z} = \{1\}$$

$$\Rightarrow B(z) = \frac{1}{1 - (z+z)} = \frac{1}{1-2z}$$

Ex) An integer composition of size n is a tuple of positive integers that sum to n.

(size 3: (3), $(2,1)$, $(1,2)$, $(1,1,1)$)

3 2+1 1+2 1+1+1

{ If order does not matter, this is a partition

Let P = class of positive integer, $|u| = u$

Let C = class of compositions

Then $C = \mathrm{SEQ}(P)$

$P = \mathcal{Z} \times \mathrm{SEQ}(P)$

$= \mathcal{Z}$

need an atom since SEQ(z) considers ε (we have $\mathbb{N}$, we want pos ints)

Thus $P(z) = \frac{z}{1-z}$ and $C(z) = \frac{1}{1-\frac{z}{1-z}}$

$= \frac{1-z}{1-2z}$

If $n \geq 1$

$$[z^n] C(z) = [z^n] \frac{1-z}{1-2z}$$

$$= [z^n] \left(\frac{1}{1-2z} - \frac{z}{1-2z} \right)$$

$$= 2^n - [z^{n-1}] \frac{1}{1-2z}$$

$$= 2^n - 2^{n-1}$$

$$= 2^{n-1} \quad \text{if } n \geq 1$$

{ Is this a must? $n \geq 1$

we write $SEQ_{\le r}(A) = \varepsilon + A + A^2 + \dots + A^r$

$$= 1 + A(z) + \dots + A(z)^r$$

$$= \frac{1 - A(z)^{r+1}}{1 - A(z)}$$

$$SEQ_{\ge r}(A) = A^r + A^{r+1} \dots$$

$$= \frac{A(z)^r}{1 - A(z)}$$

$$SEQ_{even}(A) = \varepsilon + A^2 + A^4 + A^6 \dots$$

$$= \frac{1}{1 - A(z)^2}$$

Types of Constructions

1. Iterative construction
2. Recursive construction

Ex) A (rooted planar) binary tree is either a single "root vertex"

or is a root vertex followed by (possibly empty) left and right subtrees

$B = \{\varepsilon, \bullet, \overset{z}{\bullet}, \overset{z}{\bullet}, \overset{z}{\bullet}, \overset{z}{\bullet}, \dots\}$

$B = \varepsilon + Z \times B \times B$

(ε: empty; Z: root; B: left subtree; B: right subtree)

GF:

$$\Rightarrow B(z) = 1 + z \cdot B(z)^2$$

$$\Rightarrow zB(z)^2 - B(z) + 1 = 0$$

This implies $B(z)$ one of the solutions to

$$\frac{1 \pm \sqrt{1-4z}}{2z}$$

consider $B(z) = \dfrac{1+\sqrt{1-4z}}{2z} = \dfrac{2+\cdots}{2z} = \frac{1}{z} + \cdots$

$\frac{1}{z} \Rightarrow$ not a formal power series. should be $\sum a_n z^n$

So $\frac{1-\sqrt{1+4z}}{2z} = 1 + z + 2z^2 + \dots = \sum_{n\geq 0} \frac{1}{n+1}\binom{2n}{n} z^n$

can approx # of bits needed to encode binary tree

Binary strings

we have seen how to count all binary strings $b_n = 2^n$

$$B(z) = \sum_{n\geq 0} b_n z^n = \sum_{n\geq 0} 2^n z^n = \frac{1}{1-2z}$$

Consider binary strings with restrictions

Concatenations: we write binary strings as sequences of digits without commas

$(0,1,1,0) = 0110$

- the concatenation of $\sigma = \sigma_1 \sigma_2 \dots \sigma_s$ and $\tau = \tau_1 \tau_2 \dots \tau_t$ is $\sigma\tau = \sigma_1 \sigma_2 \dots \sigma_s \tau_1 \tau_2 \dots \tau_t$
- the empty string ε has size 0 and $\varepsilon\sigma = \sigma = \sigma\varepsilon$ $\forall$ strings σ
- we write $\sigma^k = \underbrace{\sigma\sigma\sigma\dots\sigma}_{k \text{ times}}$
- we say σ is a substring of τ if σ lies within τ w/ no gaps
- The concatenation product of sets S & T of strings is the set $ST = \{\sigma\tau : \sigma\in S, \tau\in T\}$
- $S^k = \underbrace{SS\dots S}_{k \text{ times}}$

Note: $\{1,10\}\{1,01\} = \{11, \underbrace{101}_{\text{same!}}, 101, 1001\} = \{11, 101, 1001\}$ vs cartesian $\{(1,1),(10,1),(1,01),(10,01)\}$ different

- we also define $S^* = \bigcup_{k \geq 0} S^k$, $S^0 = \{\varepsilon\}$

Ex) $\{0,1\}^*$ = set of all binary strings

$\{0,1\}\{0,1\}^*$ = set of all non empty binary strings

$(\{0,1\}^2)^*$ = set of all even length binary strings

Note: A set of strings can be specified in more than one way

Ex: $\{0,1\}^*$ = $\{\varepsilon, 0, 1\}^*$ = all binary strings

can be SEQ

no GF, not SEQ

} bijection iff unambig?

we need to consider ways to produce strings, not just the sets of strings we get

Regular Expression:

An expression defined recursively by one of the following:

1. $\varepsilon, 0, 1$
2. $R \cup S$ where R and S are regular expressions
3. RS " " " " " "
4. R^* " " is a regular expression

A regular expression (R) produces a set of strings (R)

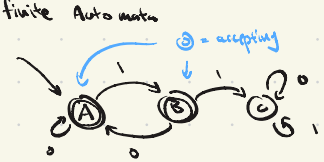
shorthand: $\{0,1\} = 0 \cup 1$

Ex)

- $(00)(1 \cup 01)$ produces $\{001, 0001\}$
- $(0 \cup 1)^*$ produces the set of all binary strings
- $(10)^5(111)$ produces $\{10101010101111\}$

The set of strings produced by a regular expression is called a regular language.

Regular Languages are important in CS because they are the accepted strings from finite Automata



all binary strings with no 2 1's in a row

a rational lof implies a finite automata exists.

Ex) $(1 \cup 10)(1 \cup 01)$

produces $\{1, 10\}\{1, 01\}$

$= \{11, 101, 101, 1001\}$

the string 101 is produced in 2 different ways

A regular expression is unambiguous if it produces each string in only 1 way. Otherwise it is ambiguous.

Ex) $(0 \cup 1)^*$ produces all strings and is unambiguous.

$(\varepsilon \cup 0 \cup 1)^*$ " " " " " ambiguous. $\neq \varepsilon 1$

How can you show an expression is unambiguous?

- $\varepsilon, 0, 1$ are unambiguous
- If R and S are unambiguous and produce R and L:
  - $R \cup S$ is unambiguous $\Leftrightarrow R \cap L = \emptyset$
  - RS is " $\Leftrightarrow$ if every $rs \in RL$ is produced by only one $r \in R$, $s \in L$
  - R^* is " $\Leftrightarrow R^k$ & R^k and R^k are disjoint

Lemma: $0^*(1\,0^*)^*$ produces all binary strings and is an unambiguous regular expression

proof Consider 00|10|101, we break up a string by starting with (possibly) some zeros, then break into ones.

we can uniquely decompose a binary string based on its location of ones

ie, $\underbrace{(0 \ldots 0)}_{m_1}\underbrace{(1 0 \ldots 0)}_{m_2} \ldots \underbrace{(1 0 \ldots 0)}_{m_r}$

for some $r \geq 0$ and $m_1, \ldots m_r \geq 0$

Ranging over all $r \geq 0$ and $m_1, \ldots, m_r \geq 0$ gives all binary strings

Corollary:

$1^*(01^*)^*$ unambiguously produces all binary string

These expressions can be modified to produce strings with restrictions

Ex) Given an unambiguous regular expression producing the set of binary strings where every zero is followed by at least one 1.

Consider the unambiguous expression $1^*(01^+)^*$. A zero can have a $\{\varepsilon, 1, 11 \ldots\}$ to follow

To ensure at least one 1, $1^*(\overbrace{011^*}^{\text{atoms}})^*$ which is ambiguous (identical to treating 01 as an atom)

$\frac{1}{1-x} \cdot \frac{1}{1-\left(\frac{x^2}{1-x}\right)}$

future: regex can be converted into generating functions

Unambiguous expressions give us generating functions; hence why ambiguity is imp.

ε	gives 1	
0 and 1	gives x	
$R \cup S$	gives $R(x)+S(x)$	{ R and S are disjoint, so sum
Concat prod. RS	leads to $R(x)S(x)$	{ if unambiguous, bijection b/t cartesian & concat product
R^*	yields $\frac{1}{1-R(x)}$	{ R^* unambiguous means no empty string in R

A block of a binary string is a nonempty maximal subsequence of the same digit

Ex)

$\underbrace{000}_{3}\ \underbrace{111}_{3}\ \underbrace{0}_{1}\ \underbrace{11}_{2}\ \underbrace{00}_{2}$

Theorem: The expressions

$$0^*(11^+ 00^+)^* 1^*$$ and $$1^*(00^+ 11^+)^* 0^*$$

blocks must be nonempty

can end on 1

unambiguously produce all binary strings

"blocks" true... how can we verify whether we've counted something correctly?

Idea: either starts w/ a 0 (or nothing) then some 1-blocks or 0-blocks, repeating

Ex) Find the generating function for the set of binary strings where every block has length 1 or 2.

we can modify the unambiguous expression $1^*(00^+ 11^+)^* 0^*$ to make all blocks ≤ 2 and get the unambiguous expression

$$(\varepsilon \cup 1 \cup 11)(0 \cup 00)(1 \cup 11)^*(\varepsilon \cup 0 \cup 00)$$

this has GF:

$$B(x) = (1+x+x^2)\left(\frac{1}{1-(x+x^2)^2}\right)(1+x+x^2)$$

$$= \frac{(1+x+x^2)^2}{(1-x-x^2)(1+x+x^2)}$$

$$= \frac{1+x+x^2}{1-x-x^2}$$

how to know what series this is? GF $\leftrightarrow$ series!

Ex) Find the GF for the set of binary strings where every block of 0s has even length.

how do we know?

A block of even length 0s can be represented by the unamby. expression

$$(00)(00)^*$$

Thus the set in question is produced by the unamby. expr

$$1^*\left((00)(00)^*11^*\right)^*(00)^*$$

with GF:

$$S(x) = \frac{1}{1-x} \cdot \frac{1}{1-\frac{x^3}{(1-x)(1-x^2)}} \cdot \frac{1}{1-x^2}$$

$$= \frac{1}{1-x-x^2}$$

Ex) Find the GF for the set of all binary strings that don't contain 011 as a substring.

starting with $1^*(00^*11^*)^*0^*$, we restrict to having at most 1 after any 0

giving the expression $1^*(00^*1)^*0^*$

and GF: $\frac{1}{1-x} \cdot \frac{1}{1-\frac{x^2}{1-x}} \cdot \frac{1}{1-x}$

$$= \frac{1}{1-2x+x^3}$$

Bivariate GFs

A GF can track objects by size. To track other objects, we need more variables.

combinatorial class with a parameter $(C, |\cdot|, x)$

is
- a set of objects C
- a size function $|\cdot| : C \to \mathbb{N}$ s.t. there are a finite set of objects of any size
- a parameter function $x : C \to \mathbb{N}$ (fix size, then params)

Bivariate GF of $(C, |\cdot|, x)$ is $C(u,z) = \sum_{\sigma \in C} u^{x(\sigma)} z^{|\sigma|}$

$$= \sum_{n \geq 0} \left(\sum_{k \geq 0} c_{k,n} u^k \right) z^n$$

$c_{k,n}$: objects in C of size n, parameter value k

Note, $C(1,z) = \sum_{\sigma \in C} z^{|\sigma|} = C(z)$

Ex) Let B be the class of binary strings with size given by length, and parameter # of 0s.

$$B(u,z) = \sum_{n \geq 0} \left(\sum_{k \geq 0} \binom{n}{k} u^k \right) z^n$$

$$= \sum_{n \geq 0} (1+u)^n z^n$$

$$= \sum_{n \geq 0} \left((1+u) z \right)^n$$

$$= \frac{1}{1 - (1+u)z}$$

not too useful, we need series $\to$ GF

In order to build GFs we add a marker M to our constructions. (when you see M, +1)

- we imagine M to have size 0, parameter value 1 } GF $z^0 u^1 = u$
- ϵ has size 0 param. value 0 } GF 1
- z has size 1 param value 0 } GF z

→ when does u turn into "u"?

Ex) If B is the class of binary strings enumerated by length and number of zeros then

$$B = \mathrm{SEQ}(z_0 \times M + z_1)$$

$$B(u,z) = \frac{1}{1 - uz - z} = \frac{1}{1 - (1+u)z}$$

Ex) Let C be the class of integer compositions. Tracking the number of 1s gives the spec.

$$C = \mathrm{SEQ}\left(zM + \overbrace{z^2 \times \mathrm{SEQ}(z)}^{z_2}\right)$$

① we track 1s.

Thus

$$C(u,z) = \frac{1}{1 - \left(zu + \frac{z^2}{1-z}\right)}$$

$$= \frac{1-z}{1 - (1+u)z + (u-1)z^2}$$

} $u=1$ gives original GF.

Ex) Find the GF for rooted planar binary trees enumerated by size (# of vertices) and number of leaves.

Let T be the class of nonempty binary trees. Then

$$T = Z + Z \times \mathcal{E} \times T + Z \times T \times \mathcal{E} + Z \times T \times T$$

Z: no children; $Z \times \mathcal{E} \times T$: left empty, right nonempty; $Z \times T \times \mathcal{E}$: right empty, left nonempty; $Z \times T \times T$: both nonempty

We can mark non-children leaves,

$$T = M \times Z + Z \times \mathcal{E} \times T + Z \times T \times \mathcal{E} + Z \times T \times T$$

$$T(u,z) = uz + 2zT(u,z) + zT(u,z)^2$$

Solving:

$$T(u,z) = \frac{1 - 2z \pm \sqrt{1 - 4z + (1-u)4z^2}}{2z}$$

remember $T(1,z) = T(z)$

when $\pm$?

$+$ has a $\frac{1}{z}$ term

$$= uz + (2u)z^3 + (u^2+4u)z^4 + \dots$$

Given $C(u,z) = \sum_{n\geq 0}\left(\sum_{k\geq 0} C_{k,n}\, u^k\right) z^n$, the formal partial derivative of C with respect to u is

$$\frac{d}{du}C(u,z) = C_u(u,z) = \sum_{n\geq 0}\left(\sum_{k\geq 0} k\, C_{k,n}\, u^{k-1}\right) z^n$$

The expected value or average of the parameter $\chi : \mathcal{C} \to \mathbb{N}$ among the objects of size n is

$$\mathbb{E}_n[\chi] = \sum_{k\geq 0} k \cdot \left[P(\chi(\sigma) = k) \text{ when } |\sigma| = n\right]$$

$$= \sum_{k\geq 0} k \cdot \left[\frac{\#\text{ objects of } \chi(\sigma) = k,\ |\sigma| = n}{\#\text{ objects with } |\sigma| = n}\right]$$

$$= \sum_{k\geq 0} k \cdot \left(\frac{C_{k,n}}{C_n}\right)$$

Prop: $\forall n \in \mathbb{N}$

$$\mathbb{E}_n[\chi] = \frac{[z^n]\, C_u(1,z)}{[z^n]\, C(1,z)}$$

proof: $[z^n]\, C_u(u,z) = \sum_{k\geq 0} k \cdot C_{k,n}\, u^{k-1}$

$$[z^n]\, C_u(1,z) = \sum_{k\geq 0} k \cdot C_{k,n}$$

Ex) Find the average number of 0s among the binary strings of length n

we know $B(u,z) = \frac{1}{1-(1+u)z}$, so $B_u(1,z) = \frac{z}{(1-(1+u)z)^2}\Big|_{u=1}$

$= \frac{z}{(1-2z)^2}$

By the negative binomial theorem implies $(n \geq 1)$

$$[z^n]\frac{z}{(1-2z)^2} = [z^{n-1}](1-2z)^{-2}$$
$$= 2^{n-1}\cdot[z^{n-1}](1-z)^{-2}$$
$$= 2^{n-1}\binom{n}{n-1}$$
$$= n\,2^{n-1}$$

So expected value is $\frac{n2^{n-1}}{2^n} = \frac{n}{2}$

Ex) The GF for compositions enumerated by size and number of 1s is

$$C(u,z) = \frac{1-z}{(1-z)(1-uz)-z^2} \quad \text{and} \quad [z^n]C(1,z) = 2^{n-1}$$

Furthermore,

$$C_u(1,z) = \frac{(1-z)^2 z}{\left((1-z)(1-uz)-z^2\right)^2}\Big|_{u=1} = \frac{(1-z)^2 z}{(1-2z)^2}$$

Then, for $n \geq 3$

$$[z^n]C_u(1,z) = [z^n]\left(\frac{z-2z^2+z^3}{(1-2z)^2}\right)$$
$$= [z^{n-1}](1-2z)^{-2} - 2[z^{n-2}](1-2z)^{-2} + [z^{n-3}](1-2z)^{-2}$$
$$= \cdots$$
$$= 2^{n-2} + 2^{n-3}\cdot n$$

Thus, if $n \geq 3$, the average is

$$\frac{2^{n-2} + 2^{n-3} n}{2^{n-1}} = \frac{2+n}{4}$$

if $n=0$, GU $=0$

$n=1$ $\quad u=1$

$n=2$ $\quad u=1$

The variance of the parameter χ on the objects of size n is

$$\mathrm{Var}_n[\chi] = \mathbb{E}_n[\chi^2] - \mathbb{E}_n[\chi]^2$$

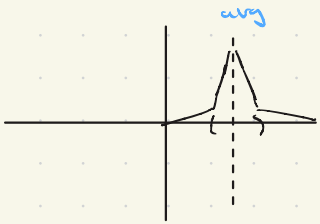
note, $\mathbb{E}_n[\chi^2] = \sum_{k \geq 0} k^2 \cdot \frac{C_{k,n}}{C_n}$ taking the second-order derivative gives k^2

prop: $$\mathrm{Var}_n[\chi] = \frac{[z^n]\left(C_{u\,u}(1,z) + C_u(1,z)\right)}{[z^n]\,C(1,z)} - \left(\frac{[z^n]\,C_u(1,z)}{[z^n]\,C(1,z)}\right)^2$$

prop: If $\mathbb{E}_n[\chi] \neq 0$ for suff. large n and $\lim_{n\to\infty}\left(\frac{\mathrm{Var}_n[\chi]}{\mathbb{E}_n[\chi]^2}\right) = 0$

then $\forall\, \varepsilon > 0$ the probability that a uniformly random object $\sigma \in \mathcal{C}$ of size n satisfies

$$(1-\varepsilon)\mathbb{E}_n(\chi) \leq \chi(\sigma) \leq (1-\varepsilon)\mathbb{E}_n[\chi] \quad \text{goes to 1 as } n \to \infty.$$



LIFT (Lagrange implicit function theorem)

Let R be a power series in z

Suppose R has no constant term and satisfies $z = \frac{R(z)}{G(R(z))}$

For some series $G(t)$ such that $\frac{1}{G(t)}$ is a power series

Then $[z^n] R(z) = \frac{1}{n} [t^{n-1}] G(t)^n$. As an extension $[z^n] R(z)^k = \frac{k}{n} [t^{n-k}] G(t)^n$

(ex) The class T of nonempty rooted planar trees consists of a root vertex followed by some (possibly none) added subtrees. It satisfies

$$T = z \times SEQ(T)$$

Thus, $T(z) = \frac{z}{1-T(z)}$, and $z = \frac{T(z)}{G(T(z))}$, where $G(t) = (1-t)^{-1}$

so

$$[z^n] T(z) = \frac{1}{n} [t^{n-1}] (1-t)^{-n} \qquad (\text{LIFT})$$

$$= \frac{1}{n} (-1)^{n-1} \binom{-n}{n-1} \qquad (\text{Negative binomial theorem})$$

$$= \frac{1}{n} \binom{2n-2}{n-1}$$

Note, $\frac{1}{n+1} \binom{2n}{n}$ is binary trees on n vertices.

Ex) Fix a positive integer r.

Find the number of rooted planar trees where every node has 0 or exactly r nonempty subtrees

This combinatorial description implies the class T of these nonempty trees satisfies $T = \mathcal{Z} + \mathcal{Z} \times T^r$

$$T(z) = z + z\,T(z)^r$$

$$= z\left[1 + T(z)^r\right]$$

Therefore, $z = \frac{T(z)}{G_T(T(z))}$ where $G_T(t) = 1 + t^r$, so LIFT and the binomial theorem imply

$$[z^n]\,T(z) = \frac{1}{n}[t^{n-1}](1+t^r)^n$$

$$n-1 = k(r) \quad \begin{cases} 0 & \text{if } r \nmid n-1 \\ \frac{1}{n}\binom{n}{n-1} & \text{if } r \mid n-1 \end{cases}$$

Ex) The class of rooted planar 5-ary trees satisfies

$$F = \mathcal{E} + \mathcal{Z} F^5$$

$$F(z) = 1 + z\,F(z)^5$$

↙ hard to LIFT

Catalan Theory!?

Let $S(z) = F(z) - 1$, then $S(z) = z[S(z)+1]^5$ { guarantees the same?!

So $z = \frac{S(z)}{G(S(z))}$ where $G(t) = (1+t)^5$, by LIFT:

for $n \geq 1$ $\quad [z^n] F(z) = [z^n] S(z)$

$$= \frac{1}{n} [t^{n-1}] (1+t)^{5n}$$

$$= \frac{1}{n} \binom{5n}{n-1}$$

$$F(z) = 1 + \sum_{n \geq 1} \frac{1}{n} \binom{5n}{n-1} z^n$$

With the knowledge of LIFT:

Suppose $F(u,z)$ is a bivariate generating function for parameter χ:

$$\mathbb{E}_n[\chi] = \frac{[z^n]\left[\frac{d}{du} F(u,z)\right]\Big|_{u=1}}{[z^n]\, F(u,z)}$$

here, $[z^n] \frac{d}{du} F(u,z)\Big|_{u=1} = \frac{d}{du} [z^n] F(u,z)\Big|_{u=1}$

Also, the bivariate GF for binary trees enumerated by size and number of leaves is

$$N(u,z) = uz + 2zN(u,z) + zN(u,z)^2$$

Thus, $z = \frac{N(u,z)}{G(N(u,z))}$ where $G(t) = u + 2t + t^2$ (u in G? allowed?)

Then $[z^n] \frac{d}{du} N(u,z)\Big|_{u=1}$

$$= \frac{d}{du}\left[[z^n] N(u,z)\right]\Big|_{u=1}$$

$$= \frac{d}{du}\left[\frac{1}{n}[t^{n-1}](u+2t+t^2)^n\right]\Big|_{u=1}$$

$$= \frac{1}{n}[t^{n-1}]\left[\frac{d}{du}(u+2t+t^2)^n\Big|_{u=1}\right]$$

$$= \frac{1}{n}[t^{n-1}]\left[n(u+2t+t^2)^{n-1}\Big|_{u=1}\right]$$

$$= [t^{n-1}](1+2t+t^2)^{n-1}$$

$$= [t^{n-1}](1+t)^{2(n-1)} = \binom{2n-3}{n-1}$$

So, $\mathbb{E}_n[X] = \dfrac{\binom{2n-2}{n-1}}{\frac{1}{n+1}\binom{2n}{n}} = (n+1)\cdot\dfrac{(2n-2)!}{(2n)!}\cdot\dfrac{n!}{(n-1)!}\cdot\dfrac{n!}{(n-1)!}$

$= \dfrac{n(n+1)}{2(2n-1)} \sim \dfrac{n}{4}$ { quarter of vertices of trees of size n are leaves.

Rational Generating functions

A C-finite sequence $(f_n) = f_0, f_1 \ldots$ is a sequence that satisfies a linear recurrence $f_n + c_1 f_{n-1} + c_2 f_{n-2} + \cdots + c_r f_{n-r} = 0$ for all $n \geq N$ where each $c_i \in \mathbb{R}$ and $c_r \neq 0$ and $N \geq r$

The integer r is the order of the recurrence

Note: the solutions to (*) are closed under addition and multiplication, which form a vector space. A particular solution is distinguished by its initial coefficients $f_0, f_1 \ldots f_{N-1}$

(ex) The fibonacci numbers satisfy

$$f_{n+2} = f_{n+1} + f_n \quad (n \geq 0), \text{ with initial conditions } f_0 = f_1 = 1$$

If we want, we can write as:

$$f_n - f_{n-1} - f_{n-2} = 0 \quad \forall\ n \geq 2$$

Theorem: (f_n) satisfies (*) if and only if $F(z) = \sum_{n \geq 0} f_n z^n = \dfrac{G(z)}{H(z)}$ where

$$G(z) = g_0 + g_1 z + \ldots + g_{N-1} z^{N-1}$$

$$H(z) = 1 + c_1 z + \ldots + c_r z^r$$

with $g_k = f_k + c_1 f_{k-1} + \ldots + c_r f_{k-r} \quad \forall\ 0 \leq k \leq N-1$ where $f_j = 0$ if $j < 0$

Proof:

Define $c_0 = 1$ so $H(z) = c_0 + c_1 z + \cdots + c_r z^r$

Then $H(z)F(z) = \left(\sum_{k=0}^{r} c_k z^k\right)\left(\sum_{k\geq 0} f_k z^k\right)$

$$= \sum_{n\geq 0}\left(\sum_{k=0}^{r} c_k f_{n-k}\right) z^n$$

0 satisfies recurrence

Thus (*) holds iff $H(z)F(z)$ is the polynomial $G(z)$ of degree $N-1$

Ex) The GF of Binary strings with no blocks of size larger than k is

$$W(z) = \frac{1 - z^{k+1}}{1 - 2z + z^{k+1}}$$

So $W_n = 2W_{n-1} - W_{n-k-1}$ $\forall$ $n \geq k+2$ { how

Ex) ℓ_n : 1, 11, 21, 1211, 111221 ... sequence

let d_n = # of digits in ℓ_n

$$D(z) = \frac{1 + z + \cdots - 12z^{78}}{1 - 2z \cdots + 6z^{76}}$$

Corollary. If (f_n) and (g_n) are C-Finite then so are (f_n+g_n) and $\left(\sum_{k=0}^{n} f_k g_{n-k}\right)$

The C-Finite Coefficient Theorem

Foundations:

- The fundamental theorem of algebra: any polynomial $P(x) \in \mathbb{C}[x]$ of degree d can be factored

$$P(x) = c(x-\lambda_1)\cdots(x-\lambda_d) \quad \text{for} \quad c, \lambda_1, \ldots, \lambda_d \in \mathbb{C}.$$

Grouping together the values of λ_i,

$$P(x) = c(x-\lambda_1)^{d_1}\cdots(x-\lambda_s)^{d_s}$$

The multiplicity of λ_i is d_i. if $d_i = 1$, λ_i is a simple root

Lemma: The multiplicity of the root λ_i is the smallest positive integer k such that

$$P(\lambda_i) = P'(\lambda_i) = \ldots = P^{(k-1)}(\lambda_i) = 0 \quad \text{and} \quad P^{(k)}(\lambda_i) \neq 0$$

Proof: If $P(x) = c(x-\lambda_1)^{d_1}\cdots(x-\lambda_s)^{d_s}$, then the product gives $P^{(j)}(x)$ as a finite sum of terms.

where $j < d_i$, each term has factor $x-\lambda_i$, so $P^{(j)}(\lambda_i) = 0$ if $j < d_i$. And

$$P^{(d_i)}(\lambda_i) = c \cdot d_i! \prod_{\ell \neq i} (\lambda_i - \lambda_\ell)^{d_\ell} \neq 0$$ (roots are distinct)

{ proof?

- Partial Fraction Decomposition

Suppose $F(z) = \frac{G(z)}{H(z)}$ with degree(G) < degree(H) and

$H(z) = C(z-\lambda_1)^{d_1} \cdots (z-\lambda_s)^{d_s}$ where λ_i are distinct and nonzero complex #s.

Then $\exists$ complex numbers $C_{(1)}^{(i)}$ such that

$$F(z) = \sum_{i=1}^{s} \left(\sum_{j=1}^{d_1} \frac{C_{(1)}^{(i)}}{(1 - z/\lambda_i)^j} \right) \quad ?$$

C-Finite Coefficient Theorem

Suppose (f_n) is a C-finite sequence with rational GF

$$F(z) = R(z) + \frac{G(z)}{H(z)}, \text{ with } R \text{ a polynomial, } \deg(G) < \deg(H), \text{ and } H(0) \neq 0$$

If $\lambda_1, \ldots, \lambda_s$ are the distinct roots of H with multiplicity $d_1, \ldots, d_s$ then

$$f_n = P_1(n)\lambda_1^{-n} + \cdots + P_s(n)\lambda_s^{-n}$$

for all $n > \deg(R)$ where $P_i(n)$ is a polynomial in n of degree at most $d_i - 1$

Ex) $F(z) = \frac{1}{3(z-1)^3(2z)^1} \iff f_n = E \cdot \left(\frac{1}{2}\right)^n + (Fn^2 + bn + H)1^n$

Proof: Let $n > \text{degree}(h)$, so $f_n = [z^n] \frac{G(z)}{H(z)}$. Partial fraction decomposition gives an expression

$$\frac{G(z)}{H(z)} = \Lambda_1(z) + \cdots + \Lambda_s(z) \quad \text{where} \quad \Lambda_i(z) = \frac{c_i^{(1)}}{1 - z/\lambda_i} + \cdots + \frac{c_i^{(d_i)}}{(1 + z/\lambda_i)^{d_i}}$$

for $c_i^{(j)} \in \mathbb{C}$. The negative binomial theorem gives (m is pos integer)

$$\begin{aligned}
[z^n](1 - z/\lambda)^{-m} &= (-\lambda)^{-n} \binom{-m}{n} \\
&= (-\lambda)^{-n} \binom{m+n-1}{m-1} \\
&= \lambda^{-n} \frac{(n+m-1)\cdots(n+1)}{(m-1)!}
\end{aligned}$$

Thus, $[z^n]\Lambda_i(z) = \lambda_i^{-n} P_i(n)$ where $P_i(n)$ is a polynomial in n of degree $\le d_i - 1$

Corollary: If (f_n) satisfies

$$f_n + c_1 f_{n-1} + \cdots + c_r f_{n-r} = 0 \quad \forall\ n \ge r, \qquad f_n = P_1(n)\lambda_1^{-n} + \cdots + P_s(n)\lambda_s^{-n} \quad \forall n \ge 0$$

where the λ_i are roots of the characteristic polynomial

$$H(z) = 1 + c_1 z + \cdots + c_r z^r$$

and $\deg(P_i) \le$ (multiplicity of λ_i as a root of H) $- 1$.

Ex) Find a closed form for the sequence f_n that satisfies $f_{n+3} = 3f_{n+1} - 2f_n$

$\forall\ n \geq 0$, $f_0 = f_1 = 4$ and $f_2 = 13$

A) the characteristic polynomial $H(z) = 1 - 3z^2 + 2z^3$ has factorization

$$H(z) = (1-z)^2(1+2z)$$

So the C-finite coefficient theorem implies $\exists\ A, B, C \in \mathbb{C}$

such that $f_n = (A + Bn)\cdot 1^{-n} + C \cdot \left(-\frac{1}{2}\right)^{-n}$

$$= A + Bn + C(-2)^n$$

from our base cases of $n = 0, 1, 2$:

$(n=0)$ $\quad 4 = f_0 = A + C$

$(n=1)$ $\quad 4 = f_1 = A + B - 2C$

$(n=2)$ $\quad 13 = f_2 = A + 2B + 4C$

$$\Rightarrow A = 3,\ B = 3,\ C = 1$$

hence $f_n = (3 + 3n) + (-2)^n$

Lemma: Assume the hypothesis of the C-finite Coefficient Theorem.

If λ_k is a simple root of $H(z)$ and $G(\lambda_k) \neq 0$ then the polynomial $P_k(n)$ in the formula for f_n is

$$P_k(n) = \frac{-G(\lambda_k)}{\lambda_k H'(\lambda_k)}$$

Since λ_k is a simple root, we can write $H(z) = (z-\lambda_k) I(z)$

for a polynomial $I(z)$ with $I(\lambda_k) \neq 0$

Partial Fraction Decomposition

$$\frac{G(z)}{H(z)} = \frac{G(z)}{(z-\lambda_k) I(z)} = \frac{C}{z-\lambda_k} + \frac{J(z)}{I(z)}$$

Clearing denominators gives

$$G(z) = C \cdot I(z) + J(z-\lambda_k)$$

setting $z = \lambda_k$, we get

$$G(\lambda_k) = C \cdot I(\lambda_k)$$

$$\Rightarrow C = \frac{G(\lambda_k)}{I(\lambda_k)}$$

Since $H(z) = (z-\lambda_k) I(z)$

$$\Rightarrow H'(z) = I(z) + (z-\lambda_k) I'(z)$$

$$H'(\lambda_k) = I(\lambda_k)$$

Thus the polynomial $P_k(n)$ is obtained from

$$[z^n]\frac{c}{z-\lambda_k} = \frac{-c}{\lambda_k}[z^n]\frac{1}{1-z/\lambda_k}$$

$$= -\frac{c}{\lambda_k}\cdot\lambda_k^{-n}$$

$$= \underbrace{-\frac{G(\lambda_k)}{\lambda_k H'(\lambda_k)}}_{P_k(n)}\cdot\lambda_k^{-n}$$

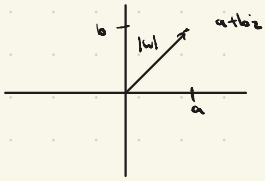
Lemma: Assume the hypotheses of the C-finite coefficient Theorem. If λ_k is a root of multiplicity m of $H(z)$ and $G(\lambda_k)\neq 0$ then the polynomial $P_k(n)$ in the expression for f_n is

$$P_k(n) = \frac{m(-1)^m G(\lambda_k)}{\lambda_k^m H^{(m)}(\lambda_k)} n^{m-1} + O(n^{m-2})$$

Let f_n and g_n be sequences of positive numbers. We write

- $f_n \sim g_n$ if $\lim_{n\to\infty}\frac{f_n}{g_n}=1$
- $f_n = O(g_n)$ if $\exists\, C>0$ and $N\in\mathbb{N}$ st $f_n \le C\cdot g_n \;\forall\, n\ge N$
- $f_n = o(g_n)$ if $\lim_{n\to\infty}\frac{f_n}{g_n}=0$

Given $w \in \mathbb{C}$ we can write $w = a+bi$



The modulus of w is $|w| = \sqrt{a^2+b^2}$

For f_n & g_n complex sequences we write

$$f_n = O(g_n) \quad \text{if} \quad |f_n| = O(|g_n|)$$

A C-finite sequence can be written

$$f_n = P_1(n)\lambda_1^{-n} + \dots + P_s(n)\lambda_s^{-n}$$

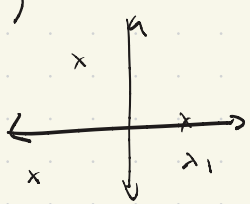
Corollary: Assume the hypothesis of the C-finite coefficient theorem.

and that the polynomials G and H are coprime.

If $|\lambda_1| < |\lambda_k| \quad \forall k \geq 2$ then

$$f_n = P_1(n)\lambda_1^{-n} + O\left(\omega^{-n} n^{\alpha-1}\right)$$

where $\omega = \min_{2 \leq k \leq s} |\lambda_k|$



?

"singularity" ?

and α is the largest multiplicity among the roots of H with modulus ω.

Corollary: under the same assumptions if $|\lambda_1| \le |\lambda_k|$ $\forall k$

and λ_1 has a higher multiplicity m, then all other roots

of H with $|\lambda_1| = |\lambda_k|$

$$f_n = \frac{m(-1)^m G(\lambda_1)}{\lambda_1^m H^{(m)}(\lambda_1)} \lambda_1^{-n} n^{m-1} \left(1 + o\left(\frac{1}{n}\right)\right)$$

Ex) The GF of the class of integer partitions where every part is

1, 2, 3 is

$F(x) = \dfrac{1}{(1-x)(1-x^2)(1-x^3)}$. Find the asymptotic behavior of partitions, n, $n \to \infty$

The denominator factors as $H(x) = (1-x)^3(1+x)(1+x+x^2)$

All roots have modulus 1 and only $x=1$ has highest multiplicity 3.

We have $G(x) = 1$, doesn't vanish at $x=1$

$$[x^n]F(x) = \left(\frac{-3}{-36}n^2 + o(n)\right) \cdot 1^n$$

$$\sim \frac{n^2}{12}$$

Vivanti-Pringsheim theorem:

Suppose $F(z) = \sum_{n \geq 0} f_n z^n = \frac{G(z)}{H(z)}$ has $f_n \geq 0 \ \forall n \geq 0$ and G and H are coprime with H not a constant, then one root of H with minimal modulus is positive and real.

Skolem's Problem:

1) does there exist an algorithm that takes a C-finite sequence (f_n) and determines whether $f_n = 0$ for any $n \in \mathbb{N}$?

2)

asymptotics $\Leftrightarrow$ C-finiteness $\Leftrightarrow$ rational relation

Combinatorial Class
- set C of objects
- size function $|\cdot| : C \to \mathbb{N}$

Finite number of objects of any size

Tools: - Bijections
- Sets

{ only / likely in class objects

Combinatorial proof

$A = B$

$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

$$= \bigcup_{k=0}^{n} B_k$$

$$= B$$

Generating Functions

C = class

C_n = objects of size n

$c_n = |C_n|$ = number of objects of size n

$$C(x) = \sum_{n \geq 0} c_n x^n$$

$$= \sum_{\sigma \in C} x^{|\sigma|}$$

$\xrightarrow[A(x)]{J}$ exists $\Leftrightarrow q_0 \neq 0$

$A(B(x))$ defined · A is polynomial
· B has no constant $b_0 = 0$

$$\frac{1}{1-x^2} = \sum (x^2)^n$$
$$= \sum x^{2n}$$

$$[x^n] \sum_{k \geq 0} a_k x^k = a_n$$

tools for coeffs:

general binom theorem

$$(1+x)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} x^n \quad \text{where} \quad \binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$$

$$(1+x)^{-k} = \sum_{n \geq 0} \binom{-k}{n} x^n = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n$$

$$\binom{n}{k} = \binom{n}{n-k}$$

$$[x^n] \frac{1+x^2}{1-x} =$$

$$[x^n]\left(A(x)+B(x)\right) = [x^n]A(x) + [x^n]B(x)$$

$$[x^n]\, x^k A(x) \begin{cases} 0 & : n < k \\ [x^{n-k}]A(x) & : n \geq k \end{cases}$$

$$[x^n]\, A(px) = p^n [x^n]\, A(x)$$

Neutral class $\mathcal{E} \Rightarrow 1$

Atomic class $\mathcal{Z} \Rightarrow z$

Sum, product, power, SEQ

$SEQ(A) \leftarrow$ no linear obj of size 0

Labelling constr.

$SEQ_{even}(A) = SEQ(A^2)$

$SEQ_{odd}(A) = A \times SEQ(A^2)$

Regular languages + Regular expressions

$\underbrace{(0 \cup 1)^*}_{\text{un ambig.}} \Rightarrow$ all bin strings $\Leftarrow \underbrace{(\varepsilon \cup 0 \cup 1)^*}_{\text{ambig.}}$

block decomp; starting 0s and 1s.

↓ can assume unambig.

LIFT $\leftarrow$ straightforward appl.

$\mathcal{C}$ finite SEQ $\in$ rational GF

$\mathcal{X}$ parameter, $\chi : \mathcal{C} \to \mathbb{N}$ review

$$C(u,z) = \sum_{\sigma \in \mathcal{C}} u^{\chi(\sigma)} \cdot z^{|\sigma|} = \sum_{n \geq 0} \left(\sum_{k \geq 0} c_{k,n} u^k \right) z^n$$

$\mathcal{C} = SEQ(\mathcal{E} \times SEQ(\mathcal{Z}))$

Averages

$\frac{\partial}{\partial u} C(u,z)\big|_{u=1}$

$$\mathbb{E}_n[X] = \frac{[z^n]\, C_u(1,z)}{[z^n]\, C(1,z)}$$

← # objects of size n

C finite $\mathbb{Q}$ ← algebraic GF

$f_{n+r} + c_{r-1} f_{n+r-1} + \dots + c_0 f_n = 0 \quad \forall n \geq 0$

$\Updownarrow$

$$F(x) = \frac{a(x)}{1 + c_1 x + \dots + c_r x^r}$$

review

$$H(x) = c\left(1 - \frac{x}{\lambda_1}\right)^{d_1} \cdots \left(1 - \frac{x}{\lambda_s}\right)^{d_s}$$

then $f_n = P_1(n)\lambda_1^{-n} + \dots + P_s(n)\lambda_s^{-n}$

$\deg(P_k) \leq$ multiplicity of $\lambda_k - 1$

$\deg(P_k) =$ mult. $- 1$ if $a(\lambda_k) \neq 0$

Asymptotics

recurrence $\Leftrightarrow$ rational GF

↘ ↙

closed form

↓

find asymptotics (unique root closest to origin)

important objects:

subsets of $[n] = \{1, \dots n\}$

2^n total

$\binom{n}{k}$ of size k

binary strings

2^n total

$\binom{n}{k}$ with k 0s / 1s

Permutations

$n!$

multisets $\binom{n+t-1}{t-1}$ of size n with t types

- Compositions 2^{n-1} of size $n \geq 1$
- Dyck paths & Motzkin
- binary (k-ary) trees

Power series from last: sufficient